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Peer Response

by Frederick Ogunlesi - Thursday, 12 February 2026, 1:58 AM

Hi Mansour,

Thanks for this well-articulated initial post. The logistics sector is one industry where we observe a significant shift in processes as we move from Industry 4.0 to 5.0. Here, we observe the augmentation of artificial intelligence in a more human-centric dynamic, prioritising employee sentiment and negative externalities. With this shift in focus away from speed and cost, it could be argued that an impact of this industrial revolution is the potential for layoffs and job insecurity due to the increased adoption of autonomous systems (Jefroy et al., 2022). How do you suggest organisations in the logistics industry balance the requirements of the shift to Industry 5.0, whilst maintaining healthy profits, and avoiding layoffs to offset costs?

I agree that the impact of any cybersecurity incident has economic and reputational impact. Despite the advancement in technology as a result of an industrial shift, these are challenges that will always be present, and as you highlighted, cybersecurity measures must be in place, as well as a clear disaster recovery plan to mitigate such risks, as poor implementation means a large economic risk for companies, especially due to the high initial investment (Kodym et al., 2020). Who should be responsible for the cost of these cybersecurity measures, and is there anything the government can do to support and incentivise companies to implement them?

References

Jefroy, N., Azarian, M., and Yu, H., 2022. Moving from Industry 4.0 to Industry 5.0: what are the implications for smart logistics?. Logistics, 6(2), p.26.

Kodym, O., Kubáč, L. and Kavka, L., 2020. Risks associated with Logistics 4.0 and their minimization using Blockchain. *Open Engineering*, *10*(1), pp.74-85.

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Peer Response

by Sarka Pekarova - Saturday, 14 February 2026, 7:29 AM

Dear Mansour,

Your post presents a clear and well-structured comparison between Industry 4.0 and Industry 5.0 within the logistics and supply chain context. I particularly agree with your observation that Industry 4.0's strong emphasis on efficiency and hyperconnectivity can unintentionally introduce systemic fragility when resilience is not developed in parallel. As Metcalf (2024) highlights, tightly integrated digital ecosystems can amplify the speed and scale at which failures propagate.

The DP World Australia ransomware incident is a compelling example because it illustrates how cyber disruption can quickly cascade through critical supply chain infrastructure. The reported halt of approximately 30,000 containers demonstrates the operational sensitivity of digitally dependent port environments and the broader economic exposure created by concentrated dependencies. From a risk engineering perspective, this incident reflects not only a cybersecurity weakness but also an architectural resilience gap that is characteristic of many Industry 4.0 implementations.

One area that could be further strengthened is the explicit socio-technical framing associated with Industry 5.0. Beyond layered cybersecurity controls and disaster recovery, Industry 5.0 emphasises human-centric resilience and trustworthy system design (European Commission, 2021). In practice, this means ensuring that critical logistics operations can degrade safely, maintain situational transparency for operators, and recover without disproportionate downstream impact.

Overall, your analysis correctly positions resilience as the key differentiator between Industry 4.0 and Industry 5.0. Expanding further on human operational continuity during cyber disruption would deepen the Industry 5.0 perspective.

References:

European Commission (2021) Industry 5.0: Towards a sustainable, human-centric and resilient European industry. Available at: https://research-and-innovation.ec.europa.eu/knowledge-publications-tools-and-data/publications/all-publications/industry-50-towards-sustainable-human-centric-and-resilient-european-industry_en (Accessed: 14 February 2026).

Metcalf, G. (2024) An introduction to Industry 5.0: History, foundations, and futures.

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